

A Further Reduction from 120 to 119 States

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1 Proof of the reduction

Let M_{120} denote the canonical 120-state machine, with working states $Q = \{0, \dots, 119\}$, initial state 0, and external halt target H . Write

$$\delta(q, a) = (b, D, r)$$

when state q , reading a , writes b , moves in direction $D \in \{L, R\}$, and enters r . All state numbers in the following reachability argument refer to M_{120} . For a configuration (q, h, τ) , the tape is $\tau: \mathbb{Z} \rightarrow \{0, 1\}$ and the head is at h . The preparation-state argument below invokes the established bootstrap and register-backend results of the original 120-state construction.

Lemma 1.1 (Preparation-state restriction). *Every visit to state 6 in the blank-tape computation of M_{120} scans zero.*

Proof. State 6 is the canonical image of the backend state `1b.reg.prep_1`. The finite bootstrap certificate in the original manuscript (the lemma titled “Bootstrap certificate”) reaches the first source decision without halting. It therefore cannot execute the completed transition $\delta(6, 1) = (1, R, H)$ during initialization.

After initialization, the register normal form and the primitive-operation argument of the original manuscript’s “Primitive register semantics” lemma apply. On entry to `1b.reg.prep_1`, the head is one cell to the right of the newly written preparation marker, in the separating zero gap. Hence the scanned cell is zero. Each primitive operation restores normal form; the control simulation and restart protocol preserve this assertion at all subsequent operations, by that manuscript’s “Backend simulation” theorem, “Natural restart of `main`” lemma, and “Dispatcher specialization” proposition. The all-tapes state quotient and the subsequent 121-to-120 projection preserve the tape and head position along the blank-tape execution, so the same restriction holds in M_{120} . This uses the established backend invariant, with no assumption about whether the source computation eventually halts. $\square$

Lemma 1.2 (Return-state restriction). *Every visit to state 37 in the blank-tape computation of M_{120} scans one.*

Proof. We give a direct argument from the transition table. Set

$$\begin{aligned} A &= \{39, 48, 74, 80, 90, 93, 94, 113, 117\}, \\ B &= \{72, 75, 102, 112\}. \end{aligned}$$

For each state $q \in A \cup B$, there is exactly one incoming transition. Its source p is listed below; in every case the instruction is $\delta(p, 1) = (1, R, q)$.

Target q	Source p	Target q	Source p
39	33	72	59
48	40	75	61
74	60	102	95
80	65	112	108
90	76		
93	81		
94	82		
113	109		
117	114		

Since none of these states is the initial state, every configuration in one of them follows such a transition. The cell immediately to the left of the new head is the cell just written. Consequently

$$q \in A \cup B \implies \tau(h-1) = 1. \quad (1)$$

The complete lists of incoming transitions to states 86 and 89 are

$$\begin{aligned} \delta(72, 1) &= (1, R, 86), \\ \delta(75, 1) &= \delta(102, 1) = \delta(112, 1) = (1, R, 89). \end{aligned}$$

Their source states belong to B . Thus (1), together with the newly written one, gives

$$q \in \{86, 89\} \implies \tau(h-2) = \tau(h-1) = 1. \quad (2)$$

The complete list of incoming transitions to state 32 is

$$\begin{aligned} \delta(29, 1) &= \delta(115, 1) = (1, R, 32), \\ \delta(86, 1) &= \delta(89, 1) = (0, L, 32). \end{aligned}$$

The right-moving transitions leave a newly written one immediately to the left of the head. For a left-moving transition, the cell immediately to the left of the new head was two cells to the left of the old head, and is one by (2). Therefore

$$q = 32 \implies \tau(h-1) = 1. \quad (3)$$

Finally, the incoming transitions to state 37 are exactly

$$\delta(p, 1) = (0, L, 37) \quad (p \in A \cup \{32\}), \quad \delta(37, 0) = (0, L, 37).$$

Every entry from a different state therefore lands on a one, by (1) and (3). If a configuration in state 37 scanning zero existed, consider the first such configuration. It is not initial and cannot arise from a different state. Its predecessor must therefore use the transition $(37, 0)$, contradicting the choice of the first occurrence. Thus $(37, 0)$ never occurs. $\square$

Theorem 1.3 (Reduction to 119 working states). *There is a deterministic binary Turing machine M_{119} with 119 working states whose blank-tape run agrees with that of M_{120} at every step in tape contents and head position, with control states related by a fixed projection. The two machines halt at exactly the same time, or both run forever.*

Proof. Define the state projection by $\pi(H) = H$ and

$$\pi(q) = \begin{cases} q, & 0 \leq q \leq 36, \\ 6, & q = 37, \\ q - 1, & 38 \leq q \leq 119. \end{cases}$$

Its image on working states is $\{0, \dots, 118\}$. For every $q \notin \{6, 37\}$ and $a \in \{0, 1\}$, define

$$\delta'(\pi(q), a) = (b, D, \pi(r)) \quad \text{when} \quad \delta(q, a) = (b, D, r).$$

At the merged state, define

$$\boxed{\delta'(6, 0) = (0, R, 8), \quad \delta'(6, 1) = (0, L, 10).}$$

These are respectively the original instructions $\delta(6, 0)$ and $\delta(37, 1)$, with their targets projected. The construction gives one instruction for each of the $119 \cdot 2 = 238$ state/symbol pairs.

By Theorems 1.1 and 1.2, every state/read pair occurring in the original blank-tape run belongs to

$$D = (Q \times \{0, 1\}) \setminus \{(6, 1), (37, 0)\}.$$

For every $(q, a) \in D$, the construction satisfies

$$\delta(q, a) = (b, D_0, r) \implies \delta'(\pi(q), a) = (b, D_0, \pi(r)). \quad (4)$$

There are exactly 238 such equations.

Induct on execution time. Initially the tapes are blank, the head positions agree, and $\pi(0) = 0$. Suppose the runs currently have identical tapes and head positions, with states q and $\pi(q)$. They read the same symbol a , and the original run uses a pair in D . Equation (4) gives identical writes and head moves, and projected successor states. Hence the same relation holds after the next step. Since $\pi(H) = H$ and no working state maps to H , entry to the halt target occurs simultaneously. $\square$

Corollary 1.4. *The machine M_{119} halts on the blank tape if and only if the Riemann hypothesis is false. It has exactly one halting transition, namely $(76, 1)$.*

Proof. The halting equivalence follows from Theorem 1.3 and the established RH equivalence of M_{120} . The two halting transitions in the original table are $(6, 1)$ and $(77, 1)$. The first is replaced at the merged state; the second is retained with state number $\pi(77) = 76$. No other transition is changed into a halting transition. $\square$

The reduction is valid for the blank-tape computation. Its two reachability restrictions, and therefore the reduction itself, do not assume the Riemann hypothesis. In particular, preservation holds even if the original machine eventually reaches its arithmetic-failure halt.